Multiple-choice questions

|!|BCEMC0160000001|!|

Which of the following combinations about the sensation and the corresponding receptor is ***incorrect***?

	Sensation	***Receptor***
A.	vision	photoreceptor
B.	hearing	mechanoreceptor
C.	pain	mechanoreceptor
D.	hot and cold	chemoreceptor

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000002|!|

Which of the following correctly shows the pathway of light entering the eye?

A. cornea → aqueous humour → lens → vitreous humour → retina
B. cornea → vitreous humour → lens → aqueous humour → retina
C. lens → aqueous humour → cornea → vitreous humour → retina
D. lens → vitreous humour → cornea → aqueous humour → retina

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000003|!|

Which of the following statements about the structure of the eye is ***incorrect***?

A. The posterior chamber of the eyeball is filled with the vitreous humour.
B. The yellow spot contains no rod cells.
C. The lens is the structure of the eye where most refraction of light occurs.
D. The choroid is richly supplied with blood vessels.

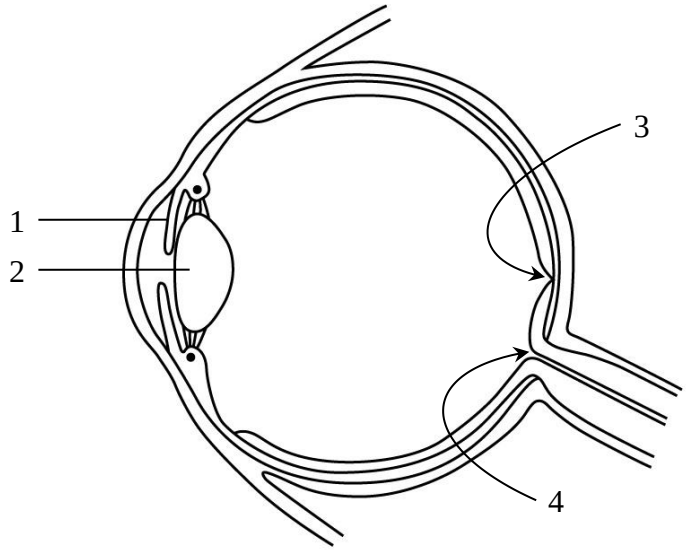
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0160000004|!|

The diagram below shows a section through the human eye.



Which structure of the eye contains a black pigment?

A. 1
B. 2
C. 3
D. 4

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000005|!|

Which of the following statements about the structure of the eye is correct?

A. There is blood supply to the cornea.
B. The choroid contains visual purple which absorbs light to reduce the reflection of light within the eye.
C. The iris muscles control the amount of light entering the eye.
D. The suspensory ligaments contract and relax to regulate the thickness of the lens.

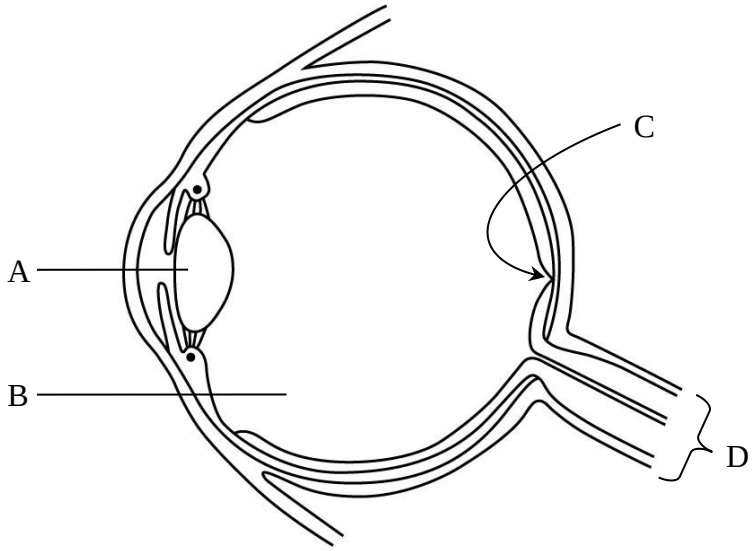
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0160000006|!|

The following diagram shows a section through the human eye.



Which of the following statements is ***incorrect***?

A. Structure A is made up of living cells without nuclei.

B. Structure B can refract light.

C. Structure C contains no photoreceptors.

D. Structure D transmits nerve impulses from the retina to the cerebrum.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0160000007|!| ★

What happens in the eye when a person enters a dark room from bright light?

A. The circular muscles of the iris contract and the radial muscles relax.

B. The circular muscles of the iris relax and the radial muscles contract.

C. The ciliary muscles contract.

D. The ciliary muscles relax.

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0160000008|!| ★

When a person enters a dark room from bright light, he is not able to see clearly for a while because

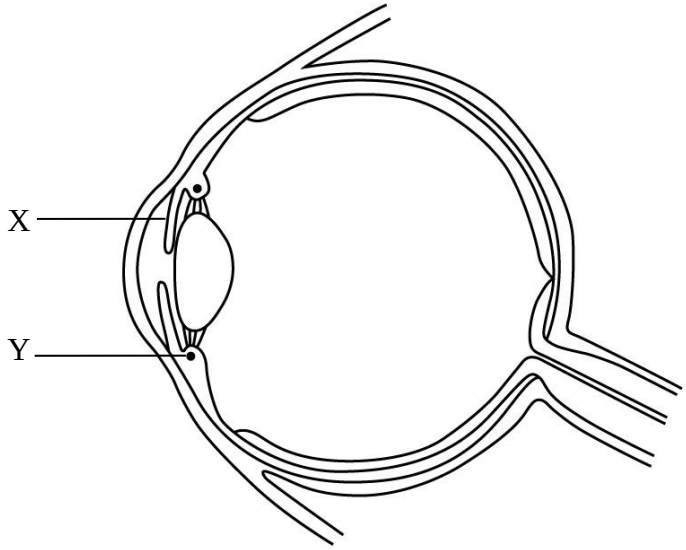
A. the retina is insensitive temporarily.
B. it takes time for the lens to adjust its thickness.
C. it takes time for the pupil to constrict.
D. it takes time for the pupil to dilate.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0160000009|!| ★

The diagram below shows a section through the human eye.



What happens to the circular muscles of X and Y when the eye focuses on a distant object in bright light?

	Circular muscles of X	***Circular muscles of Y***
A.	Contract	Contract
B.	Contract	Relax
C.	Relax	Contract
D.	Relax	Relax

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0160000010|!| ★ **Cross topic**

Some events involved in the pupil reflex are list below.

(1) The pupil constricts.
(2) Bright light shines into the eye.
(3) Nerve impulses are sent to the brain.
(4) Photoreceptors are stimulated.
(5) Nerve impulses are sent to the muscles of the iris.

Which of the following shows the correct sequence of the events?

A. (2), (4), (3), (5), (1)
B. (2), (4), (5), (3), (1)
C. (4), (3), (5), (1), (2)
D. (4), (5), (3), (1), (2)

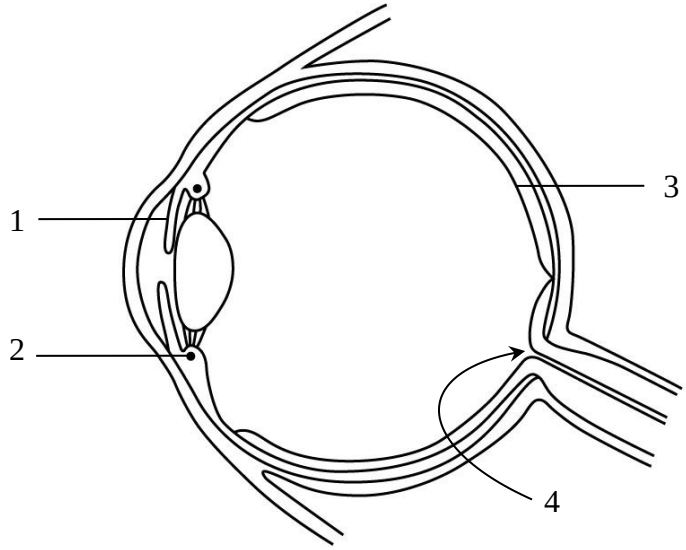
□

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000011|!| ★ **Cross topic**

The diagram below shows a section through the human eye.



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Which of the following correctly identify the sites of the receptors and effectors involved the pupil reflex?

	Receptors	***Effectors***
A.	3	1
B.	3	2
C.	4	1
D.	4	2

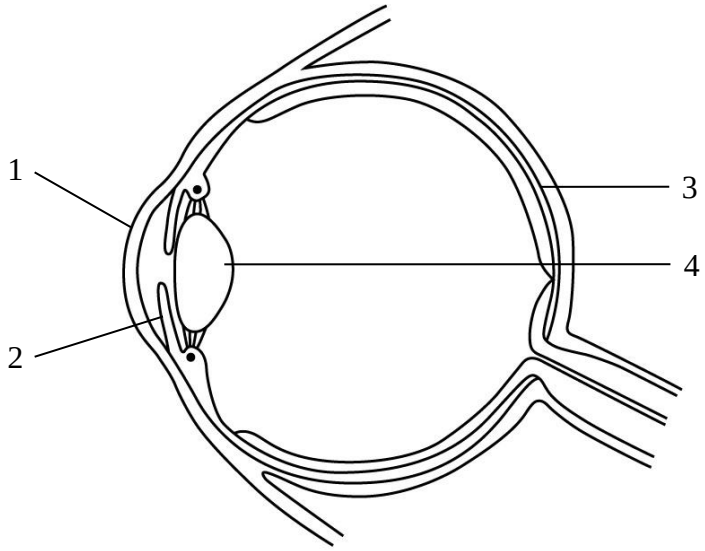
□

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0160000012|!|

The diagram below shows a section through the human eye.



Which structure gives the eye its colour?

A. 1
B. 2
C. 3
D. 4

□

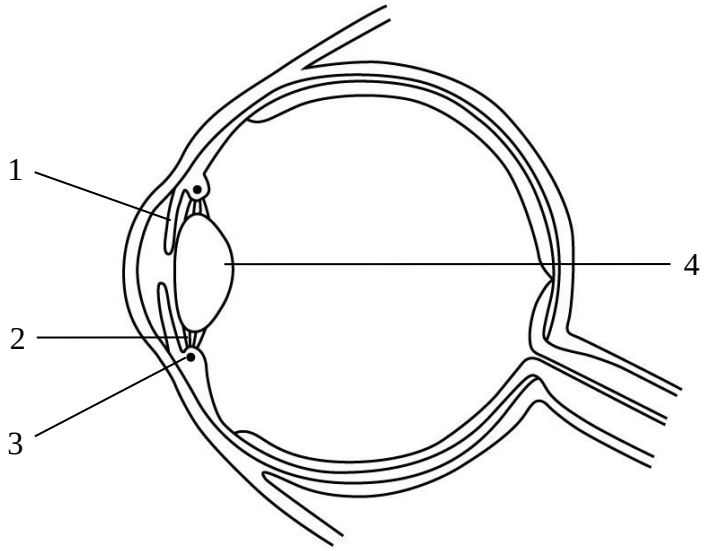
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0160000013|!|

The diagram below shows a section through the human eye.



Which of the labelled structures contain muscles?

A. 1 and 2

B. 1 and 3

C. 2 and 3

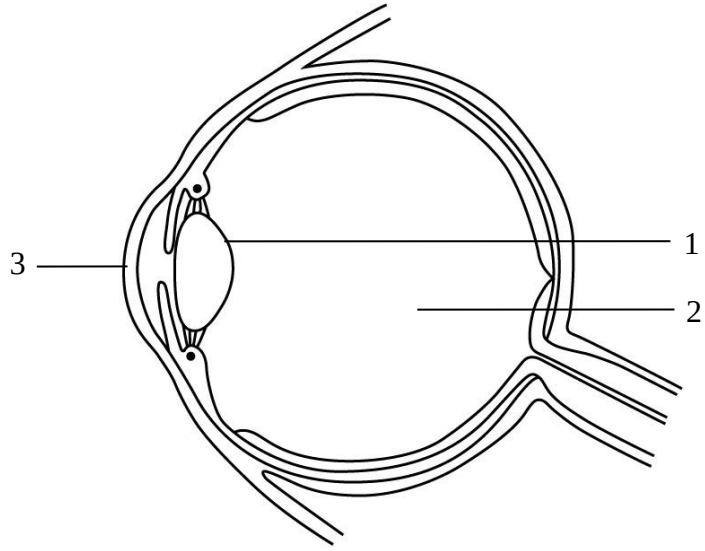
D. 3 and 4

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000014|!|

The diagram below shows a section through the human eye.



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Refraction of light occurs at

A. 1 only.
B. 1 and 2 only.
C. 1 and 3 only.
D. 1, 2 and 3.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000015|!|

Which of the following statements about the blind spot in the eye is correct?

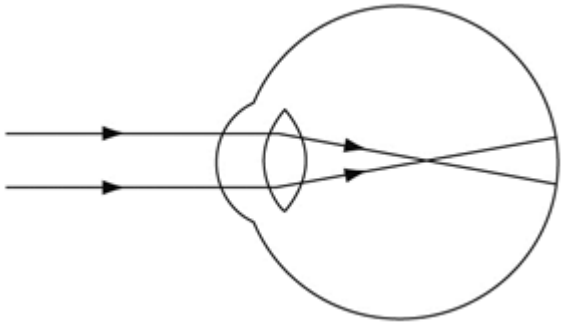
A. It contains only rod cells.
B. It contains only cone cells.
C. The image formed there is inverted.
D. No image can be formed there.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000016|!|

What condition is represented by the ray diagram below?



A. a short-sighted person looking at a nearby object
B a short-sighted person looking at a distant object
C. a long-sighted person looking at a nearby object
D. a long-sighted person looking at a distant object

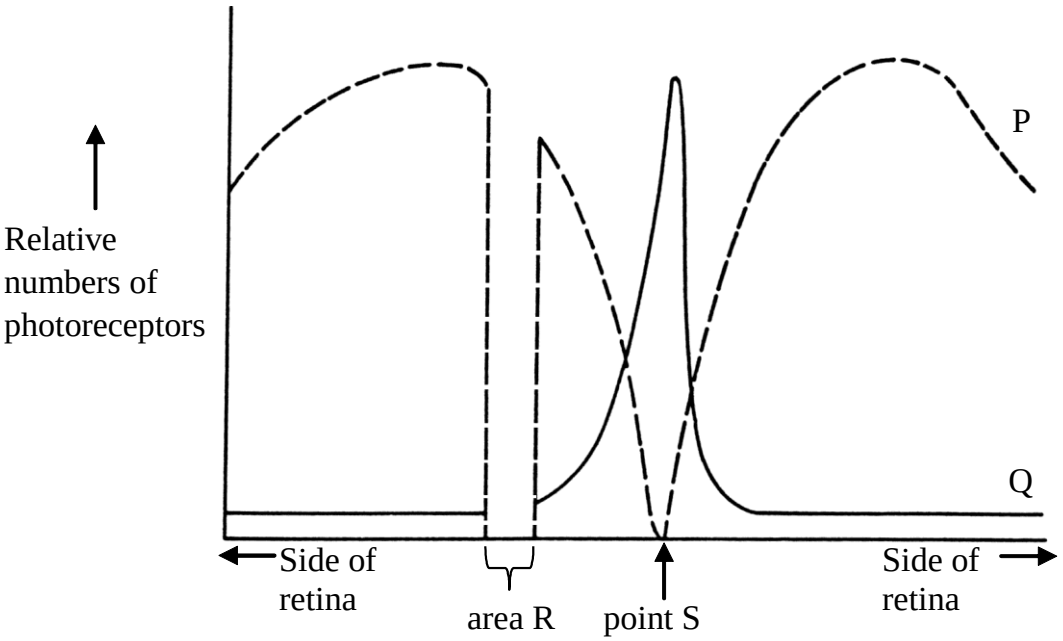
##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0160000017|!|

The graph below shows the relative numbers of two types of photoreceptors, P and Q, along the horizontal section of the human retina.



Which of the following combinations correctly identify P to S?

	P	***Q***	***R***	***S***
A.	rod cells	cone cells	blind spot	yellow spot
B.	cone cells	rod cells	yellow spot	blind spot
C.	rod cells	cone cells	yellow spot	blind spot
D.	cone cells	rod cells	blind spot	yellow spot

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0160000018|!|

Which of the following statements about visual purple are correct?

(1) Visual purple is found in rod cells.
(2) Visual purple is important to night vision.
(3) Vitamin D is necessary for the formation of visual purple.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000020|!|

Which part of the human eye helps prevent the reflection of light inside the eye?

A. vitreous humour
B. sclera
C. choroid
D. retina

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000021|!|

What is the function of the choroid?

A. It protects the eyeball.
B. It supplies nutrients to the eyeball.
C. It reflects light.
D. It generates nerve impulses.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0160000023|!| ★

During a check-up, the optician may place eye drops in the patient's eye to keep the pupil dilating. Which muscles are contract when the eye drops are applied?

A. circular muscles of the iris
B. radial muscles of the iris
C. ciliary muscles
D. muscles that move the eyeball

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0160000024|!|

Which of the following combinations correctly states the condition of the ciliary muscles and the suspensory ligaments when the eye is focusing on a near object?

	Ciliary muscles	*Suspensory ligaments*
A.	contract	tightened
B.	contract	slackened
C.	relax	tightened
D.	relax	slackened

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000026|!|

Which of the following combinations correctly states the condition of the suspensory ligaments and the lens when the eye is focusing on a distant object?

	Suspensory ligaments	*Lens*
A.	slackened	became thicker
B.	slackened	became thinner
C.	tightened	became thicker
D.	tightened	became thinner

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0160000027|!| ★

When the eye is focusing on a distant object, the lens becomes less convex. This is brought about by

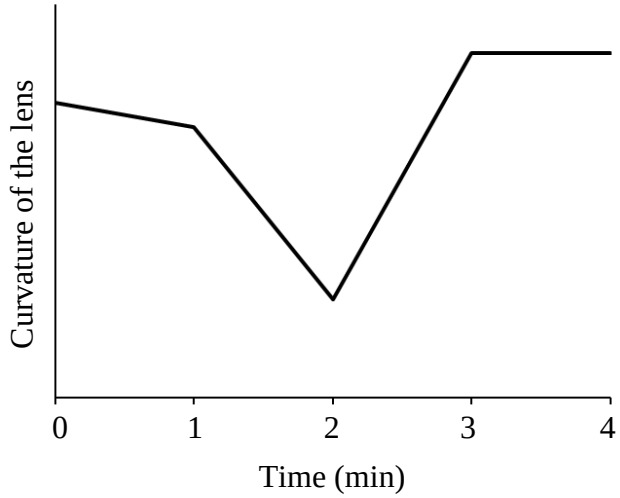
A. contraction of the suspensory ligaments.
B. relaxation of the suspensory ligaments.
C. contraction of the ciliary muscles.
D. relaxation of the ciliary muscles.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000028|!| ★

The graph below shows the changes in the curvature of the lens of a man's eye as he watches a bird.



During which period was the bird moving towards the man?

A. 0–1 min
B. 1–2 min
C. 2–3 min
D. 3–4 min

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMA0160000030|!| ★★

Peter noticed that there was a cockroach in a dim room. He found that he could see the cockroach more clearly when he tried to focus on the rubbish bin right next to it. Which of the following ***best*** explains this?

A. The image of the cockroach was formed on the yellow spot where there are cone cells only.
B. The image of the cockroach was formed on the yellow spot where there are rod cells only.
C. The image of the cockroach was formed on the periphery of the retina where there are more cone cells than rod cells.
D. The image of the cockroach was formed on the periphery of the retina where there are more rod cells than cone cells.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0160000031|!|

Which of the following structures of the eye help(s) refracting light to the retina?

(1) lens
(2) cornea
(3) aqueous humour

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0160000032|!|

Long-sighted people cannot see near objects clearly. Which of the following is/are the cause(s) of this eye defect?

(1) The lens is too thin.
(2) The eyeball is too short.
(3) The surface of the cornea is uneven.

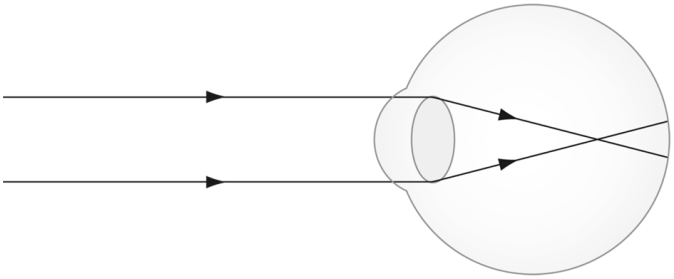
A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000033|!| ★

Below is a ray diagram of a common eye defect.



Which of the following combinations correctly identifies the eye defect and the type of lens to be worn to correct it?

	Eye defect	*Type of lens*
A.	Short-sightedness	Concave lens
B.	Short-sightedness	Convex lens
C.	Long-sightedness	Concave lens
D.	Long-sightedness	Convex lens

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0160000034|!| ★

Which of the following combinations correctly matches the problem of long-sightedness and its correction?

	Problem	***Correction***
A.	image focused in front of the retina	wear concave lens
B.	image focused in front of the retina	wear convex lens
C.	image focused behind the retina	wear concave lens
D.	image focused behind the retina	wear convex lens

□

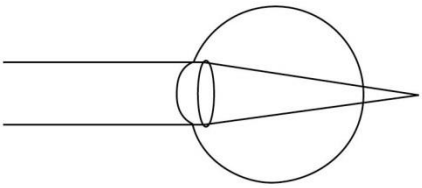
##D##

Skill assessed	Mark
Understanding basic concepts	1

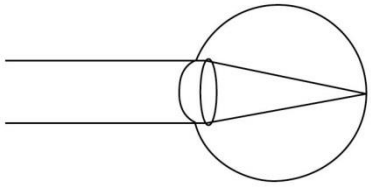
|!|BCEMB0160000035|!| ★

Which of the following diagrams represents the short-sighted condition?

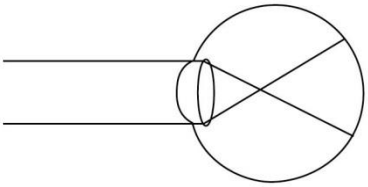
A.



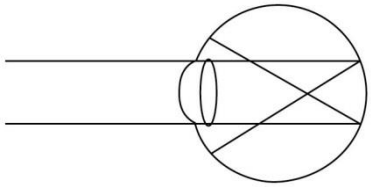
B.



C.



D.



□

##C##

Skill assessed	Mark
Analysing data	1

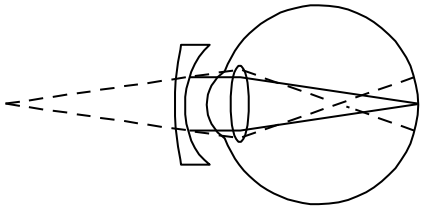
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|!|BCEMA0160000036|!| ★★

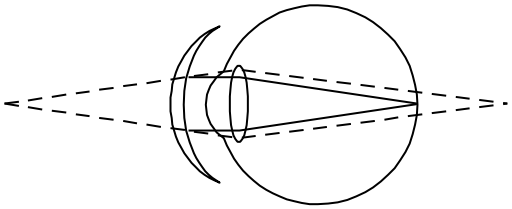
Which of the following diagrams shows the eye defect being corrected by a suitable lens?

(The broken lines represent the light rays before correction by the lens, while the solid lines represent the light rays after correction.)

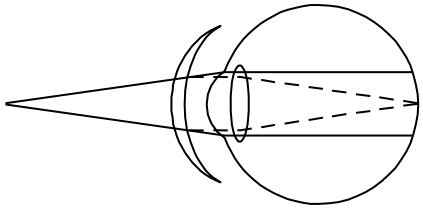
A.



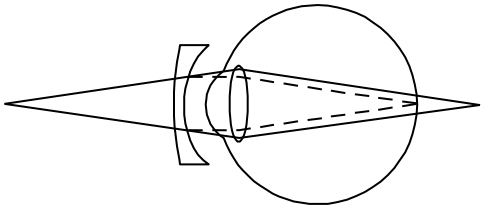
B.



C.



D.



##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0160000037|!|

Which of the following statements about colour blindness is/are correct?

(1) It is more common in men than in women.
(2) Most people with colour blindness cannot distinguish between red and green colours.
(3) It is caused by a deficiency in vitamin A.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0160000038|!| ★

A person suffers from night blindness and has difficulty in focusing on distant objects. Which of the following can explain his conditions?

A. insufficient amount of visual purple; eye ball too long
B. insufficient amount of visual purple; eye ball too short
C. insufficient cone cells in the retina; eye ball too long
D. insufficient cone cells in the retina; eye ball too short

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0160000039|!| ★

Which of the following correctly explains why our eyes feel tired after staring at a computer screen for a long time?

A. The lens has remained in its convex state for a long time.
B. The suspensory ligament has been tightened for a long time.
C. The ciliary muscles have contracted for a long time.
D. The photoreceptors have been over-stimulated.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0160000041|!|

Which of the following statements about the structure of the ear is ***incorrect***?

A. The ear drum separates the outer ear from the middle ear.
B. The middle ear is liquid-filled.
C. The round and oval windows separate the middle ear from the inner ear.
D. The Eustachian tube is filled with air.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0160000042|!|

Which of following shows the correct route of the transmission of sound waves / vibrations in the ear?

A. pinna → Eustachian tube → eardrum → ear bones → oval window
B. pinna → Eustachian tube → eardrum → ear bones → round window
C. pinna → auditory canal → eardrum → ear bones → oval window
D. pinna → auditory canal → eardrum → ear bones → round window

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000043|!|

Which of the following combinations correctly matches the ear structure with its function?

	Ear structure	***Function***
A.	pinna	protecting the ear
B.	auditory canal	equalizing the air pressure on both sides of the eardrum
C.	eardrum	amplifying sound waves
D.	ear bones	transmitting vibrations

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000044|!|

Which of the following statements about the process of hearing is ***incorrect***?

A. The pinna collects sound waves and directs them along the auditory canal.
B. The ear bones transmit and amplify the vibrations from the eardrum to the oval window.
C. Vibrations from the round window cause the perilymph and the endolymph in the cochlea to vibrate.
D. Nerve impulses are transmitted to the brain along the auditory nerve.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0160000045|!|

Which part of the human ear is concerned with body balance?

A. ear bones
B. semicircular canals
C. cochlea
D. Eustachian tube

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000046|!|

Which part(s) of the human ear contain(s) sensory hair cells?

(1) cochlea
(2) semicircular canals
(3) eardrum

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000047|!| ★

Which of the following is/are the function(s) of the inner ear?

(1) detecting sound vibrations
(2) controlling body balance
(3) detecting body movement

A. (1) only
B. (3) only
C. (1) and (3) only
D. (1), (2) and (3)

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000050|!|

Which of the following statements about auxins are correct?

(1) They are produced at shoot tips and root tips.
(2) They promote cell division.
(3) They promote cell elongation.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000052|!| ★

How do auxins cause plant shoots to grow towards unilateral light?

A. increasing cell division on the illuminated side of the shoot
B. increasing cell division on the shaded side of the shoot
C. increasing cell elongation on the illuminated side of the shoot
D. increasing cell elongation on the shaded side of the shoot

□

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0160000053|!|

When a plant shoot bends towards sunlight, what process(s) do/does auxins promote in the cells on the shaded side of the shoot?

(1) mitosis
(2) cell differentiation
(3) cell elongation

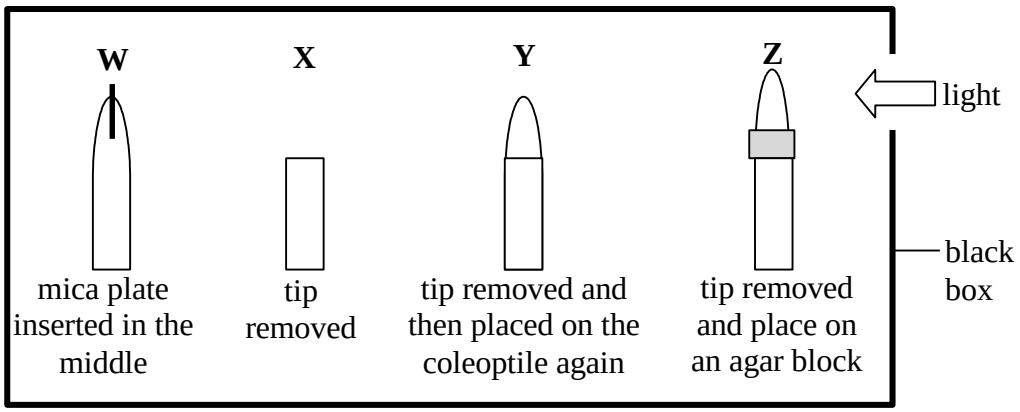
A. (1) only
B. (3) only
C. (1) and (2) only
D. (1), (2) and (3) only

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000054|!| ★ SI

The diagram below shows a set-up to study the effect of light on the growth of oat coleoptiles.



Which coleoptile(s) will bend towards the light source?

A. Y only
B. Z only
C. Y and Z only
D. W, Y and Z only

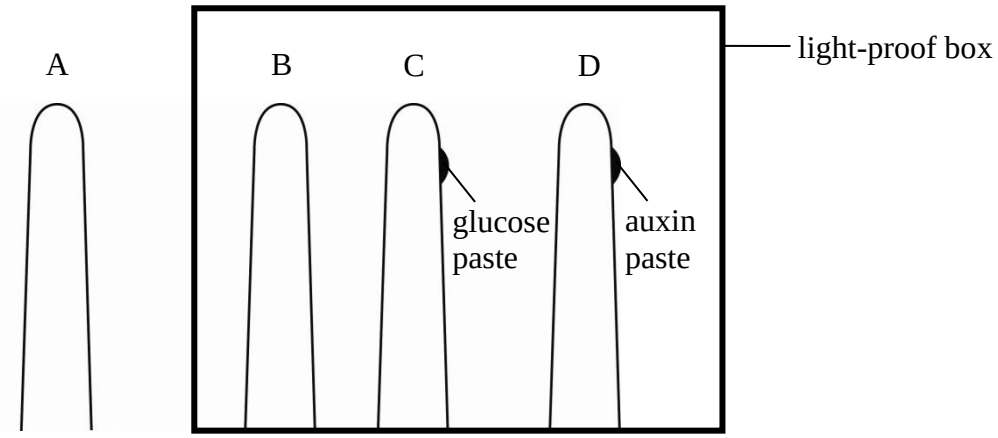
##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0160000055|!| ★ **SI**

Which of the following coleoptile will bend towards one side?



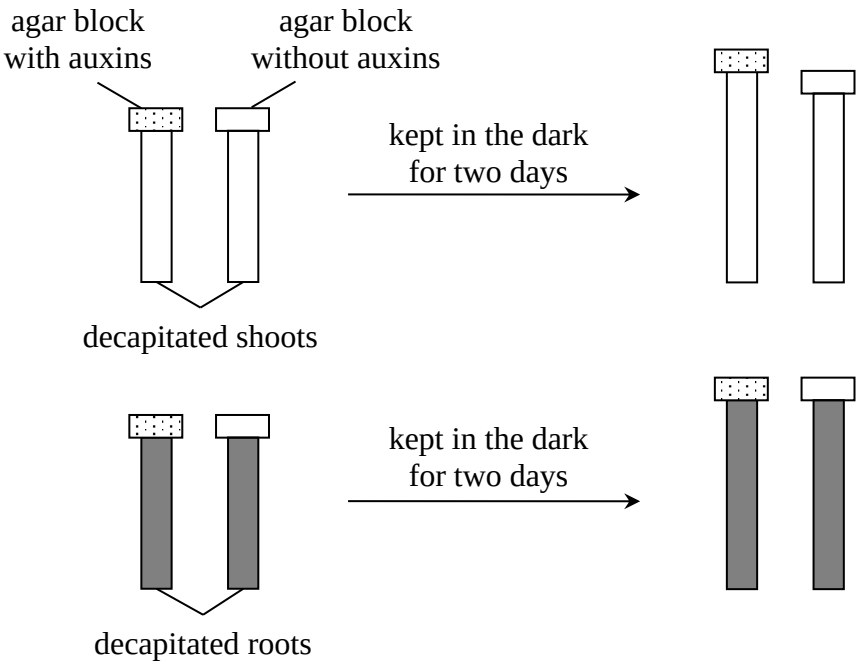
□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0160000056|!| ★ **SI**

The diagram below shows the set-ups of an experiment and the results after two days.



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What effects does this concentration of auxins have on the growth of the shoot and the root?

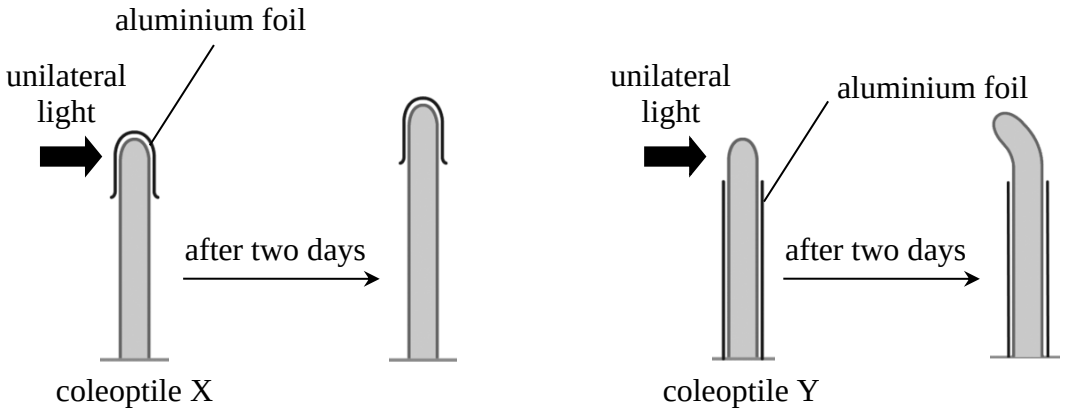
	Shoot	***Root***
A.	growth promoted	growth promoted
B.	growth not promoted	growth inhibited
C.	growth promoted	growth not promoted
D.	growth inhibited	growth not promoted

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0160000057|!| ★★ **SI**

The diagram below shows the set-ups of an experiment and the results after two days.



What conclusion can be drawn from this experiment?

A. Auxins are responsible for causing phototropic response.
B. Auxins are produced at the tip.
C. The tip is essential for growth.
D. The tip is sensitive to unilateral light.

##D##

Skill assessed	Mark
Practical skills	1

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|!|BCEMC0160000058|!|

Which of the following statements about a coleoptile exposed to unilateral light is correct?

A. More auxins are produced on the illuminated side than on the shaded side.
B. More auxins are produced on the shaded side than on the illuminated side.
C. Auxins are transported from the illuminated side to the shaded side.
D. Auxins are transported from the shaded side to the illuminated side.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000059|!|

When an oat coleoptile is exposed to unilateral light,

(1) more auxins are produced on the shaded side.
(2) auxins move from the illuminated side to the shaded side.
(3) the illuminated side grows slower than the shaded side.

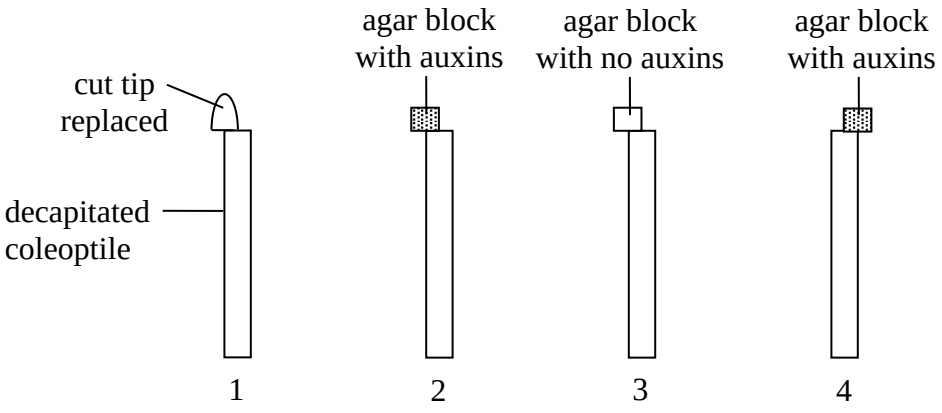
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000060|!| ★ **SI**

The diagram below shows coleoptiles treated in different ways.



Which two coleoptiles can be used to test the hypothesis that auxins are growth-promoting substances in coleoptiles?

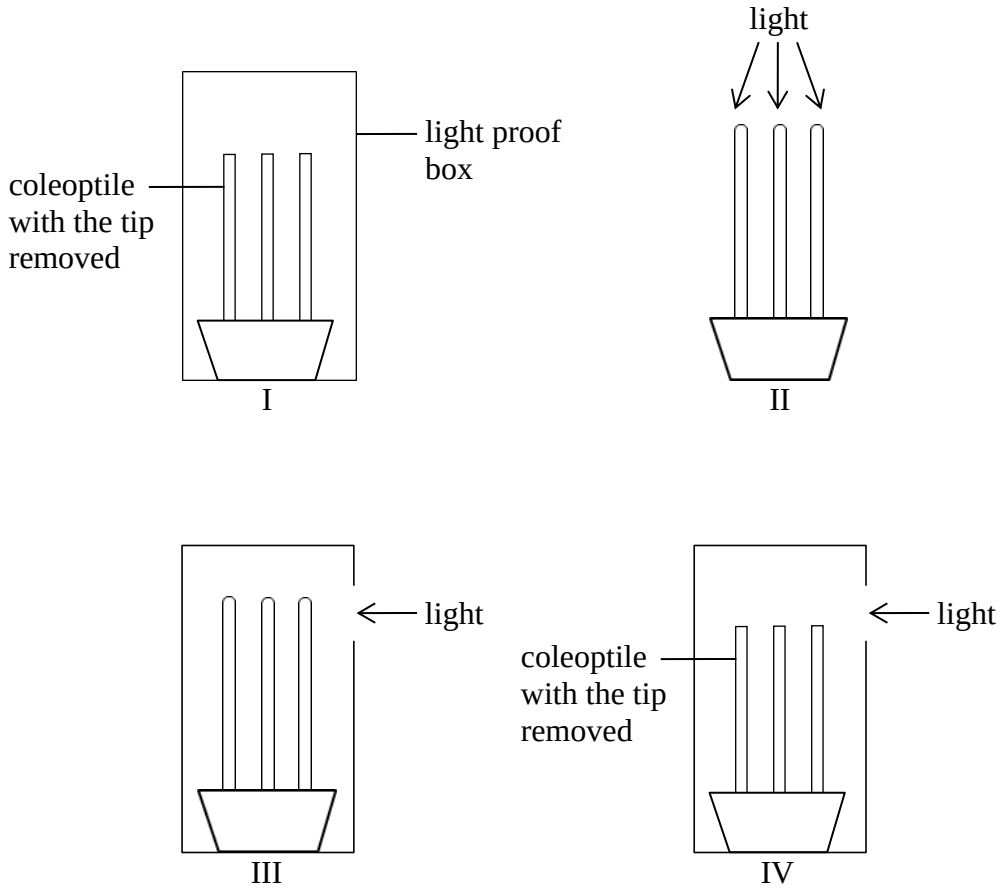
A. 1 and 2
B. 2 and 3
C. 2 and 4
D. 3 and 4

##B##

Skill assessed	Mark
Practical skills	1

|!|BCEMB0160000061|!| ★ **SI**

The diagram below shows an investigation studying phototropic responses in oat coleoptiles.



Which two set-ups can be used to show that the shoots of the coleoptiles are positively phototropic?

A. I and II
B. I and III
C. II and III
D. II and IV

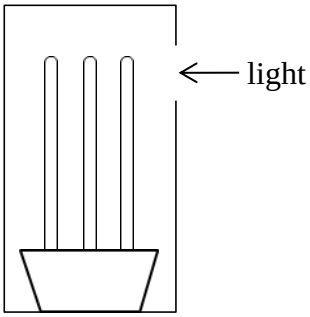
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##C##

Skill assessed	Mark
Practical skills	1

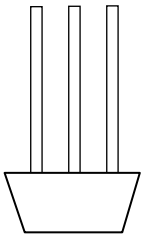
|!|BCEMB0160000062|!| ★ **SI**

The diagram below shows an experimental set-up used to study phototropic responses in coleoptiles.

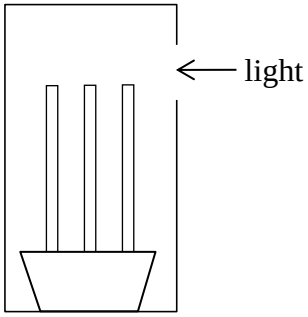


Which of the following can be used as a control set-up to test the hypothesis that the tip is sensitive to light?

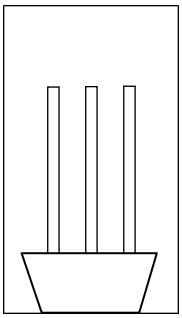
A.



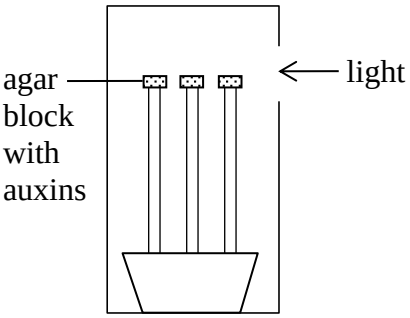
B.



C.



D.



□

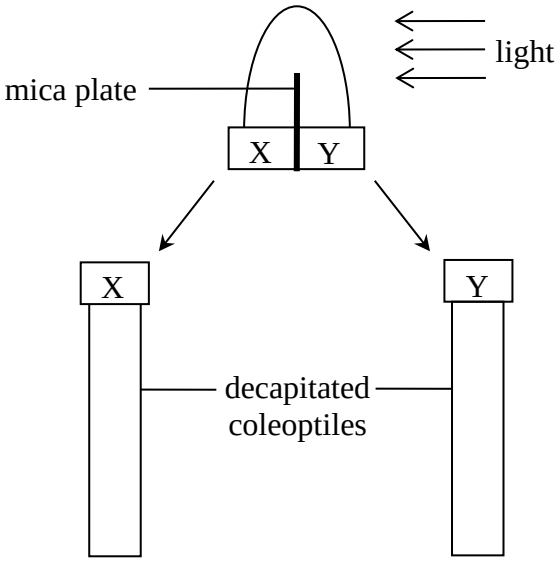
##B##

Skill assessed	Mark
Practical skills	1

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|!|BCEMB0160000064|!| ★ **SI**

The diagram below shows a cut tip of an oat coleoptile which was placed on two agar blocks, X and Y, separated by a mica plate. The set-up was illuminated from one side. Agar blocks X and Y were then placed on top of two decapitated coleoptiles, P and Q, respectively. The two coleoptiles were kept in the dark for two days.



What would be expected results?

A. P would bend to the right, but Q would not.
B. Q would bend the right, but P would not.
C. P would grow longer than Q.
D. Q would grow longer than P.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0160000066|!|

Which of the following statements about tropism in plants is/are correct?

(1) It is controlled by plant hormones called auxins.
(2) It is a response to a unilateral stimulus by a plant.
(3) It usually results in growth towards the stimulus.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

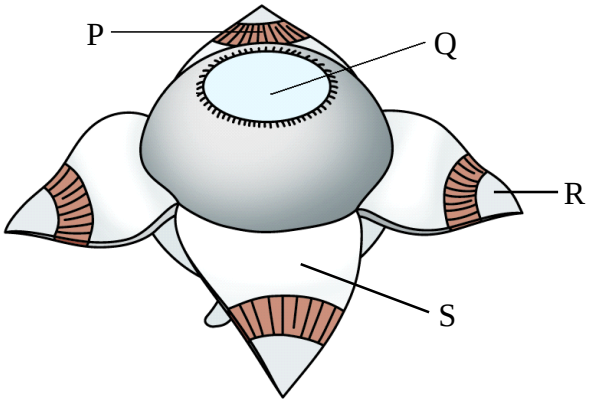
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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000067|!|

Directions: The following two questions refer to the diagrams below, which shows a dissected ox eye.



Which of the labelled structures is the most difficult to be cut through when dissecting the eye?

A. P
B. Q
C. R
D. S

☐

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000068|!|

Which of the labelled structures are transparent?

A. P and Q
B. P and S
C. Q and R
D. R and S

☐

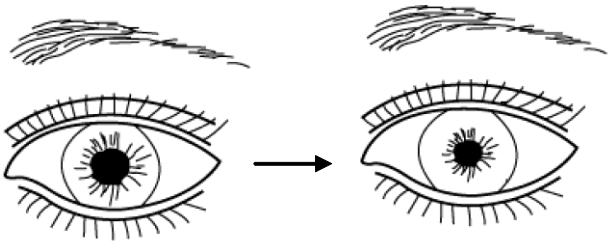
##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0160000069|!|

Directions: The following two questions refer to the diagram below, which shows the response of person's eye after a change in external conditions.



X **Y**

What change would cause this response?

A. a decrease in light intensity
B. an increase in light intensity
C. focusing on a nearby object
D. focusing on a distant object

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0160000070|!|

What changes take place in the eye from X to Y?

	Pupil	***Radial muscles in iris***	***Circular muscles in iris***
A.	dilates	contract	relax
B.	dilates	relax	contract
C.	constricts	contract	relax
D.	constricts	relax	contract

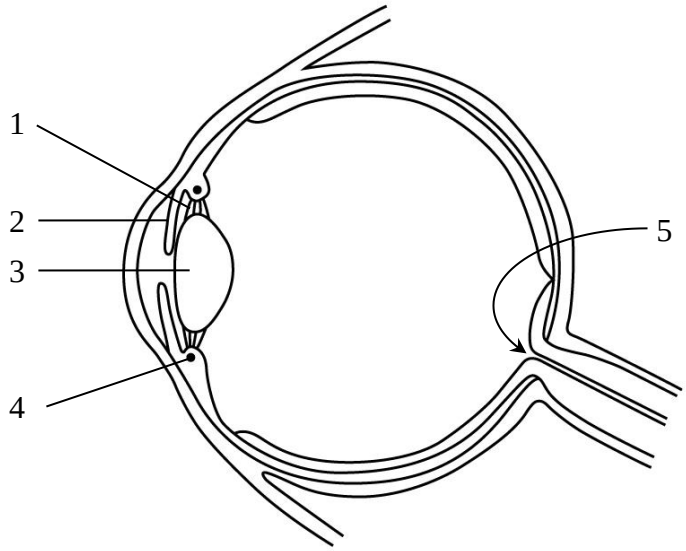
##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0160000071|!| ★

Directions: The following three questions refer to the diagram below which shows a section through the human eye.



Which structure contains the muscles that contract in response to changes in light intensity?

A. 1
B. 2
C. 3
D. 4

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000072|!| ★

Which structure contains the muscles that contract in order to produce a focused image on the retina?

A. 1
B. 2
C. 3
D. 4

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0160000073|!| ★

When an image falls on structure 5, one would see

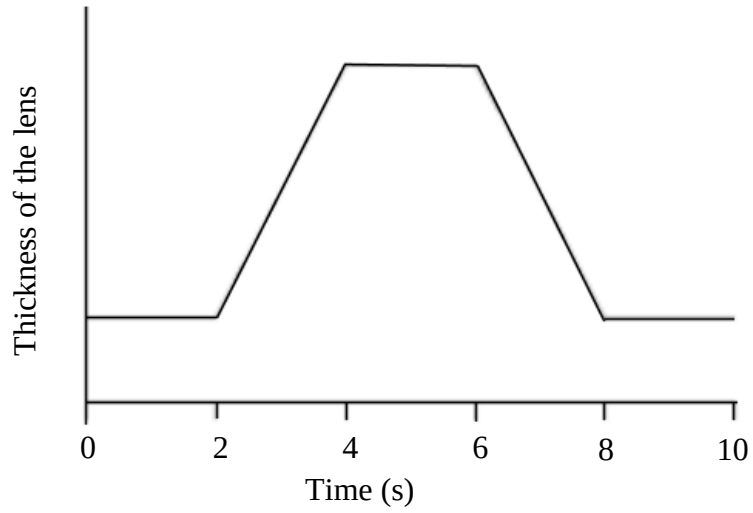
A. a blurred, black and white image because it contains rod cells only.
B. a detailed coloured image because it contains a large number of cone cells.
C. a detailed, black and white image because it contains a large number of rod cells and only a few cone cells.
D. no image at all because it does not contain any photoreceptors.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000074|!| ★

Directions: The following two questions refer to the graph below, which shows the changes in the lens thickness of a man's eye within 10 seconds.



What was happening in the man's eyes between the 2nd and 4th seconds?

A. The pupil is dilating.
B. The ciliary muscles are contracting.
C. The suspensory ligaments are contracting.
D. The suspensory ligaments are relaxing.

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0160000075|!| ★

Between the 6th and the 8th seconds, the man was looking at

A. a nearby stationary object.
B. a far away stationary object.
C. an object moving towards him.
D. an object moving away from him.

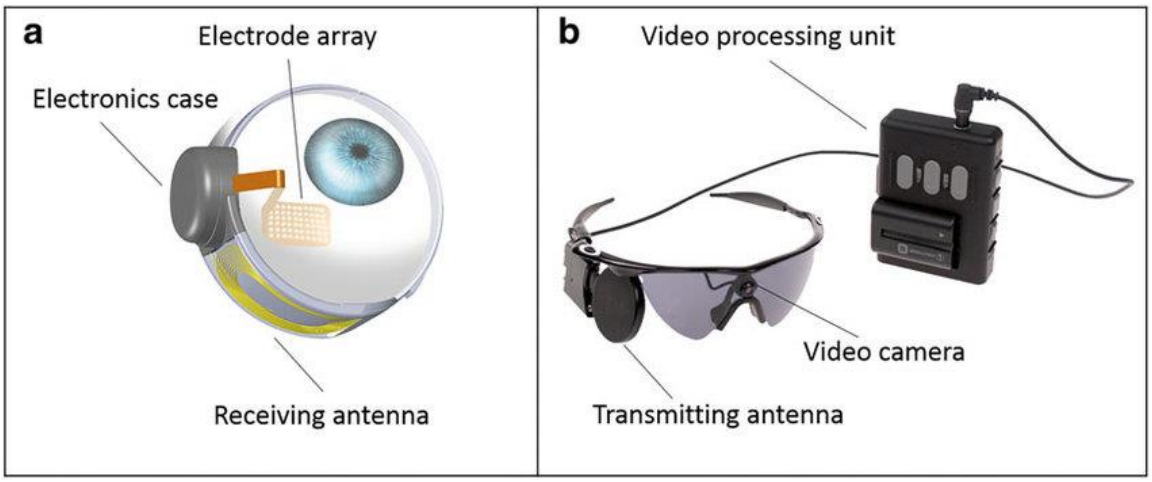
□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0160000076|!| ★★ **STSE**

Directions: The following two questions refer to an artificial retina system which helps restore limited visually impaired people. Figure A below shows the implanted components of the system. Figure B shows the external components of the system.



The video camera on the glasses captures images and sends information to the video processing unit, which converts the information to electrical signals. The signals are then transmitted to an electrode array implanted behind the retina. The retinal implant emits pulses which travel through the optic nerve to the brain. The brain then perceives patterns of light and dark.

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The artificial retina can help restore limited vision in which of the following conditions?

(1) Photoreceptors are damaged.
(2) The optic nerve is damaged.
(3) The visual cortex is damaged.

A. (1) only
B. (1) and (2) only.
C. (1) and (3) only.
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMA0160000077|!| ★★ **STSE**

Which of the following is ***not*** a limitation of the artificial retina system?

A. It does not enable users to perceive colours.
B. It does not enable users to perceive movement.
C. The user needs to learn to interpret the visual patterns generated by the system.
D. The eyeball must be present for installation of the artificial retina.

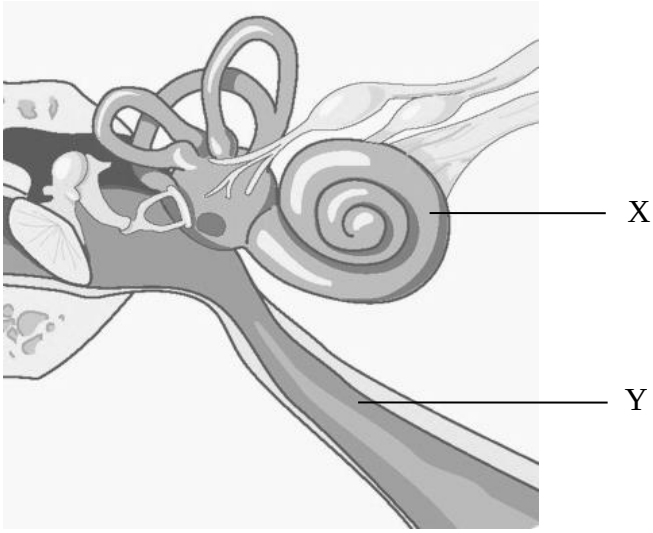
##B##

Skill assessed	Mark
Critical thinking	1

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|!|BCEMB0160000078|!| ★

Directions: The following two questions refer to the diagram below, which shows the middle and inner ears.



What is/are found in X?

(1) sensory hair cells
(2) perilymph
(3) air

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1) and (3) only

☐

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0160000079|!| ★

During take-off, Y may not function properly and the person may feel pain in the ears. Which of following may relief the condition?

A. using ear plugs
B. yawning
C. keeping the mouth closed
D. stretching the ear pinna

☐

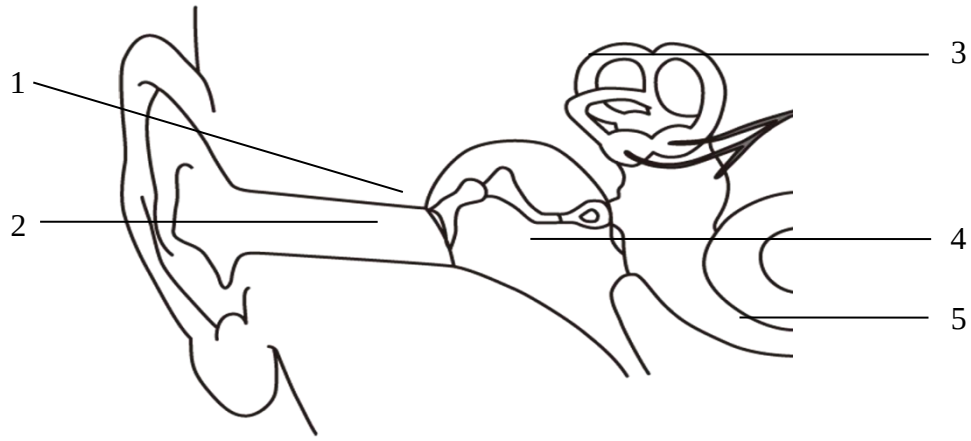
##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0160000080|!|

Directions: The following two questions refer to the diagram below, which shows the structure of a human ear.



Which of the labelled structures will vibrate when there are sound waves?

A. 1 and 2 only
B. 2 and 4 only
C. 1, 2 and 4 only
D. 1, 2 and 5 only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0160000081|!|

What is the function of structure 3?

A. to control body balance
B. to detect head movements
C. to detect vibrations
D. to transmit vibrations to structure 4

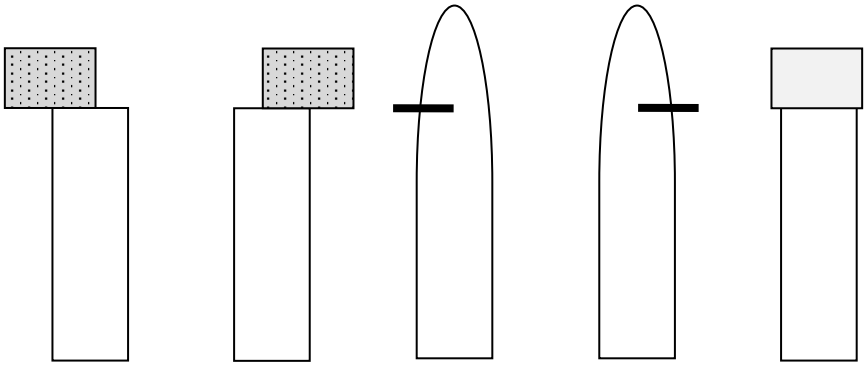
##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0160000082|!| ★ **SI**

Directions: The following two questions refer to the following experiment. Five coleoptiles, treated as shown below, were placed in darkness for one day. The tips of coleoptiles (1), (2) and (5) were cut away.



Key:

Agar block with auxins

Mica plate

Plain gar block (without auxins)

Which coleoptiles would bend towards the left?

A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0160000083|!| ★ **SI**

Which of the following statements about coleoptile (5) is correct?

A. It grows longer and bends to the right.
B. It grows longer and bends to the left.
C. It grows straight upwards.
D. It does not show any curvature or increase in length.

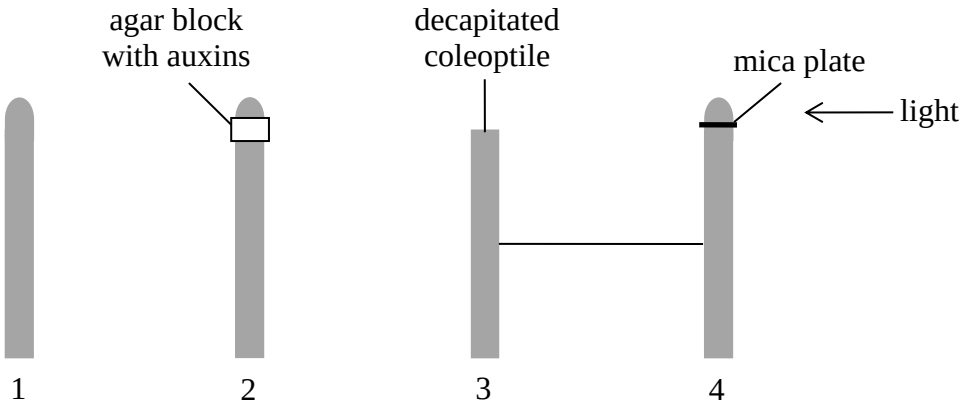
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##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0160000084|!| ★ **SI**

Directions: The following two questions refer to the diagram, below which shows four oat coleoptiles under unilateral light.



Which coleoptile(s) would bend towards the light source?

A. 1 only
B. 1 and 2 only
C. 1 and 4 only
D. 1, 2 and 4 only

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0160000085|!| ★ **SI**

Which coleoptiles can be used to show that there are growth-promoting substances passing from the tip to the growing region of the coleoptile?

A. 1, 2 and 3
B. 1, 2 and 4
C. 1, 3 and 4
D. 2, 3 and 4

□

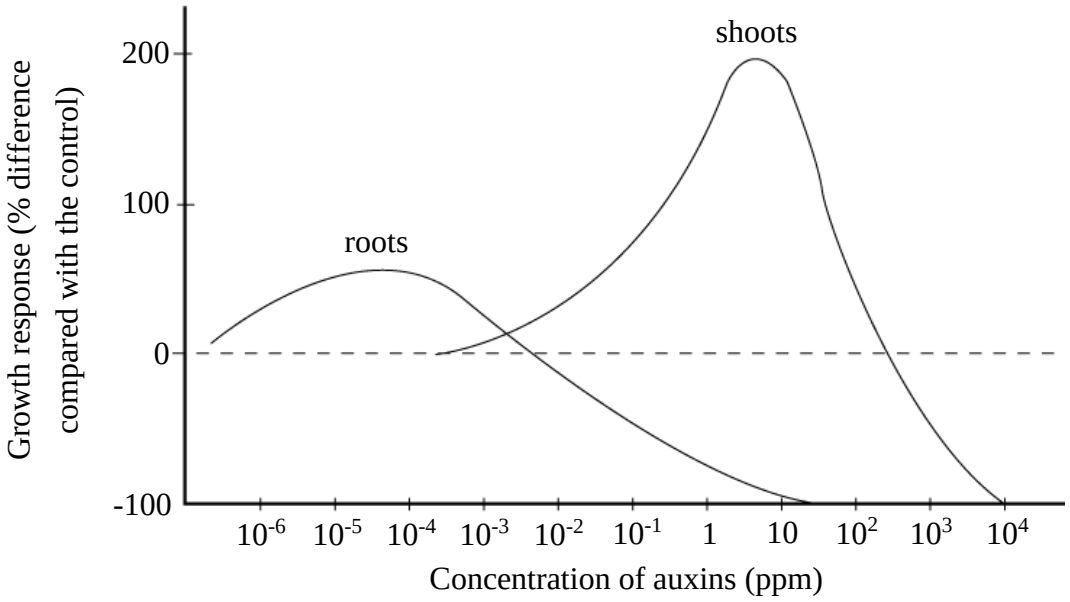
##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0160000086|!| ★

Directions: The following two questions refer to the graph below which shows the growth response of shoots and roots to different concentrations of auxins.



Which of the following correctly shows the effects of different concentrations of auxins on the growth of shoots and roots?

	Concentration of auxins (ppm)	***shoots***	***roots***
A.	10^{-5}	growth inhibited	growth promoted
B.	10^{-4}	growth not promoted	growth promoted
C.	10^{-3}	growth promoted	growth inhibited
D.	10^{3}	growth not promoted	growth not promoted

□

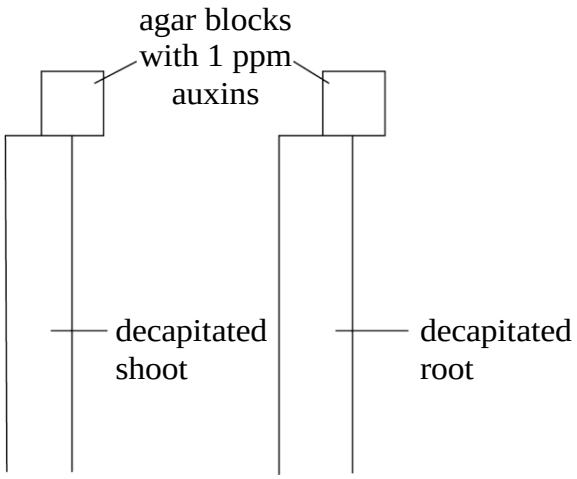
##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0160000087|!| ★

Two agar blocks each of which contained 1 ppm auxins were placed on top of one side of a decapitated shoot and a decapitated root respectively.



Which of the following combinations show the results of the decapitated shoot and root?

	Decapitated shoot	***Decapitated root***
A.	bent to the right	bent to the right
B.	bent to the right	bent to the left
C.	bent to the left	bent to the right
D.	bent to the left	bent to the left

□

##C##

Skill assessed	Mark
Analysing data	1

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